

Python 90 Days Challenge

DAY 01 — PYTHON FUNDAMENTALS

1. INTRODUCTION TO PYTHON

Python is a high-level, general-purpose programming language used for automation, web development, data handling, AI, ML, scripting, testing, and backend development.

Python has simple and readable syntax, making it beginner-friendly. Python source files use the **.py** extension.

Example:

```
print("Hello, World!")
```

Why Learn Python?

- Beginner-friendly and easy to learn.
- Simple and readable syntax.
- Large collection of libraries.
- Useful for automation, AI, data science, and backend development.
- Used across many industries.

2. HISTORY OF PYTHON

- Python was created by **Guido van Rossum**.
- Development started in the **late 1980s**.
- Python was first released in **1991**.
- It was designed to be simple, readable, and easy to use.
- The name was inspired by **Monty Python's Flying Circus**.

Period	Event
Late 1980s	Python development started
1991	First Python release
2000	Python 2.0 released
2008	Python 3.0 released

For modern learning, Python 3 is used. Version can be checked with **python --version** or **python -V**.

3. FEATURES OF PYTHON

Feature	Meaning
Easy syntax	Simple and readable
High-level	No direct machine-level details required
Interpreted	Runs using an interpreter
Dynamically typed	No manual type declaration required
Object-oriented	Supports classes and objects
Cross-platform	Works on Windows, macOS, and Linux
Large library support	Supports built-in and external libraries
Open-source	Free to use

4. ADVANTAGES & DISADVANTAGES

Advantages

- Easy to learn and read.
- Simple syntax.
- Large number of libraries.
- Useful for automation and many industries.
- Free and open-source.

Disadvantages

- Interpreted execution is usually slower than compiled execution.
- The source PDF does not provide a separate detailed list of disadvantages.

5. APPLICATIONS OF PYTHON

- Web Development
- Automation
- Data Science
- Artificial Intelligence (AI)
- Machine Learning (ML)
- Testing
- Scripting
- Backend Development

6. PYTHON INSTALLATION & ENVIRONMENT

Step 1 — Download Python

- Download a Python 3 version from the official Python website.

Step 2 — Install Python on Windows

- Run the Python installer.
- Select **Add python.exe to PATH**.
- Click **Install Now**.

Step 3 — Verify Installation

- Open Command Prompt / Terminal and run:

```
python --version
python -V
```

Step 4 — Install VS Code

- Install Visual Studio Code.
- Install the official **Python extension**.

Step 5 — Create a Python File

- Create a file with the **.py** extension, for example **hello.py**.

```
print("Hello, World!")
```

Step 6 — Run the Python Program

- Use the Run button in VS Code, or run the file from the terminal.

```
python hello.py
```

Step 7 — Python Interpreter / REPL

- The Python interpreter executes Python code.
- REPL means **Read, Evaluate, Print, Loop**.

Step 8 — Virtual Environment (venv)

- A virtual environment isolates project packages from other projects.

Step 9 — pip

- pip is Python's package installer.

Step 10 — requirements.txt

- A requirements.txt file can list the packages required by a project.

7. PYTHON SYNTAX

Python syntax is the set of rules used to write Python programs correctly.

First Python Program:

```
print("Hello, World!")
```

- **print()** is used to display output.
- Text is written inside quotation marks.
- Python programs are commonly saved in **.py** files.

Indentation

- Indentation defines blocks of code in Python.
- The common style is **4 spaces** for one indentation level.
- Avoid mixing tabs and spaces.

```
if True: print("Hello")
```

- Incorrect syntax can produce **SyntaxError** or **IndentationError**.

QUICK REVISION

- Python is a high-level, general-purpose language.
- Created by Guido van Rossum; first released in 1991.
- Python is easy, interpreted, dynamically typed, object-oriented, cross-platform, and open-source.
- Python is used in web development, automation, data science, AI/ML, testing, scripting, and backend development.
- Basic environment: Python + VS Code + Terminal + pip + venv.
- Python syntax uses indentation to define code blocks.